

Hydraulic bearing metamodel

Application to a turbine health indicator

OpenTURNS User's Day.

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Acknowledgments: Nicolas Tardieu, Amine Hassini, Elias Fekhari, Joseph Muré.

Physical model and probabilistic model

Simple Monte Carlo design of experiments

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References



Industrial context

- ▶ Industrial asset management for hydraulic production equipment.
- ▶ Equipment monitoring to improve diagnosis and prognosis.
- ▶ Create a turbine health indicator to identify the degradation state and detect malfunctions.
- ▶ Tools: OpenTURNS (see [Baudin et al., 2016]), Persalys, Salome-Meca

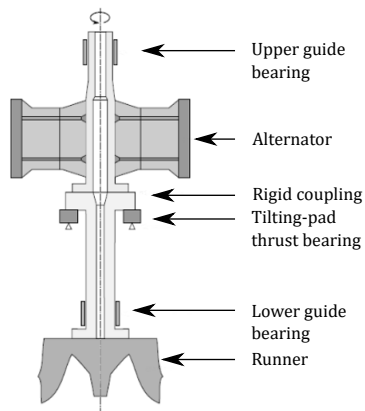


Figure 1: Hydraulic turbine (source: (Clément & Tardieu, 2021)).

Industrial context and LEGOS model

Calculation parameters:

- ▶ 3D spatial discretization: 16 000 cells.
- ▶ Unit evaluation time: approximately 1 minute and 30 seconds on a workstation.

Physical model:

- ▶ Calculation of pressure and temperature on a 3D spatial field.
- ▶ Constant tolerance: 10^{-9} .
- ▶ Outputs $P_{\max}$ and $T_{\max}$ recorded on the spatial field.

Variables of interest:

- ▶ $P_{\max}$ bar $\in]1, +\infty[$.
- ▶ $T_{\max}$ °C $\in]23, +\infty[$.

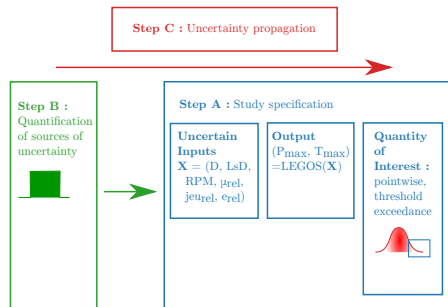


Figure 2: Generic uncertainty quantification methodology applied to the LEGOS model.

Discretization and Geometric Synthesis

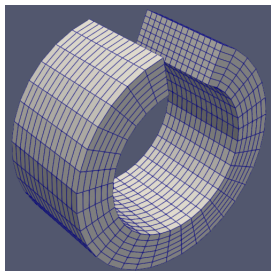


Figure 3: Cartesian representation of a finite element mesh for the simulation of hydrodynamic lubrication. The discretization is performed in cylindrical coordinates with $n_x = 32$, $n_y = 6$, and $n_z = 14$ elements.

⚠: This is not exactly the discretization used in the study.

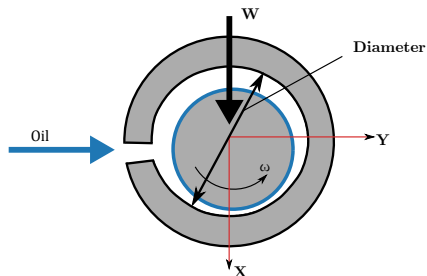


Figure 4: Bearing (source: (Hassini, Corus & Zhang, 2022), figure 13 page 22, with modifications).

Probabilistic model

Probabilistic model

- Uniform random variables and their parameters reflect the diversity of machines in the hydropower fleet¹.
- Inputs are **independent**.

Variable	Description	Distribution
D	Bearing diameter (m)	$\mathcal{U}(0.3, 0.9)$
LsD	Ratio L/D where L is the bearing length and D is the diameter (m)	$\mathcal{U}(0.4, 2.5)$
ω	Rotational speed (revolutions per minute or RPM)	$\mathcal{U}(100, 500)$
μ_{rel}	Relative viscosity of the lubricant	$\mathcal{U}(0.25, 4)$
jeu_{rel}	Relative clearance, i.e., the difference between the shaft diameter and the bearing diameter	$\mathcal{U}(0.25, 4)$
e_{rel}	Relative eccentricity	$\mathcal{U}(0.1, 0.85)$

Table 1: Description and distributions of the model's input variables.

¹See [Baudin and Clément, 2023].

Simple Monte Carlo design of experiments

Design of experiments

- ▶ Type: Simple Monte Carlo.
- ▶ Sizes: 1 000, 2 000, 5 000.
- ▶ Occasional convergence failure of LEGOS.
- ▶ Without a method to resolve LEGOS evaluation failures, these points are ignored.

Size	Successes (%)	Failures (%)
1 000	98.3 %	1.7 %
2 000	98.2 %	1.8 %
5 000	98.5 %	1.5 %

Table 2: Success and failure rates for simple Monte Carlo designs of experiments.

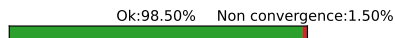


Figure 5: Proportion of successful and failed calculations in the simple Monte Carlo design of experiments³ of size $n = 5\,000$. LEGOS calculations are performed with a fixed tolerance of 10^{-9} .

Output vector distribution

The distributions of $P_{\max}$ and $T_{\max}$ are highly skewed

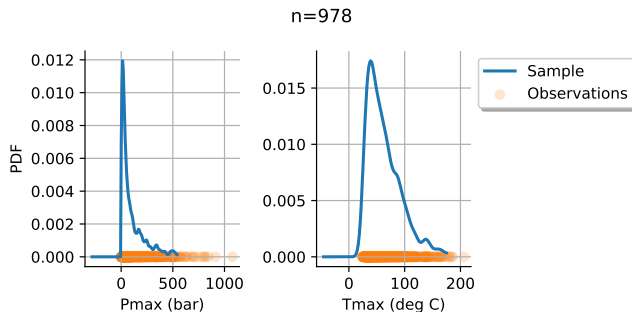


Figure 6: Marginal distribution of the output sample, estimated by kernel density estimation. Left: distribution of the maximum pressure output from the LEGOS model. Right: distribution of the maximum temperature output from the LEGOS model. LEGOS calculations use a fixed tolerance of 10^{-9} .

LEGOS non-convergences

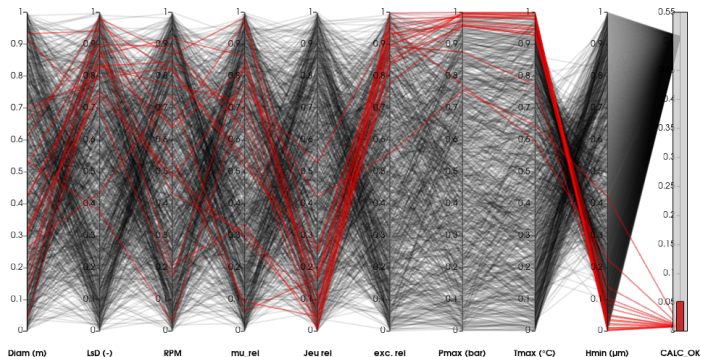


Figure 7: Analysis of LEGOS evaluation failures in the rank space. LEGOS calculations use a fixed tolerance of 10^{-9} .

Conclusion. The variables most influential on the lack of convergence of LEGOS are (the *value* indicated in parentheses more frequently leads to non-convergence):

1. Eccentricity (high),
2. Clearance (low),
3. LsD (high).

Non-convergences and evaluation failures

Failure locations:

- ▶ Exc close to 0.85 (max), LsD close to 2.5 (max), and Clearance close to 0.25 (min).
- ▶ *Domain*: Exc $\in [0.1, 0.85]$, LsD $\in [0.4, 2.5]$, Clearance $\in [0.25, 4]$.
- ▶ The inputs leading to failures are dependent.

Reduction of evaluation failures:

- ▶ The maximum failure-free domain is obtained by limiting the eccentricity to **0.63** (instead of 0.85). This approach was **abandoned** because it excessively reduces the usefulness of the metamodel.
- ▶ **Increase LEGOS tolerance.** Gradually transitioning from 10^{-9} (“fixed tolerance”) to 10^{-4} (“dynamic tolerance”) helps reduce evaluation failures.

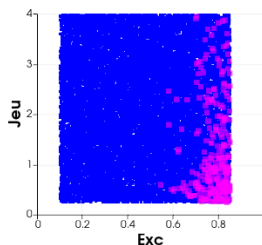


Figure 8: Points without convergence (fixed tolerance of 10^{-9}) on a Monte Carlo sample of size 20 000 in the (Exc, Clearance) plane. Blue points: correct (19,705). Pink points: without convergence (295).

Sensitivity analysis based on a threshold

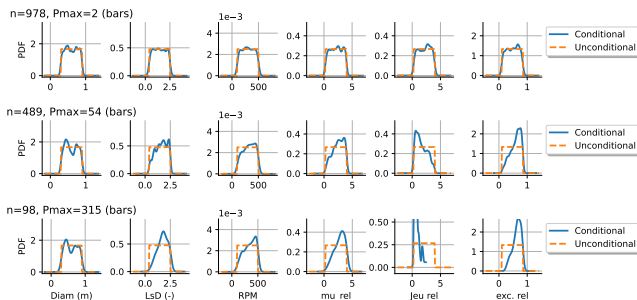


Figure 9: For each input variable, unconditional (orange, dashed line) and conditional (blue, solid line) distribution leading to $P_{\max}$ higher than a given quantile level, for different quantile levels. Top, for a quantile level equal to 0 (all simulations). Middle, for a quantile level equal to 0.5 (half of the simulations). Bottom, for a quantile level equal to 0.9 (10% of the simulations). LEGOS calculations use a fixed tolerance of 10^{-9} .

Conclusion. The input variables that more frequently lead to high pressure:

1. Clearance (low), 2. Eccentricity (high), 3. Mu (high).

Number of coefficients as a function of the polynomial degree

Creation of a metamodel

- ▶ Bearing metamodel using a polynomial chaos method.
- ▶ Variables are uniform: Legendre orthogonal polynomials.
- ▶ Coefficient estimation: by least squares or integration.
- ▶ Creation and evaluation of a larger simple Monte Carlo design of experiments: $n = 20\,000$.
- ▶ Degree selection by maximizing the Q^2 score estimated by k -fold analytical cross-validation⁴.

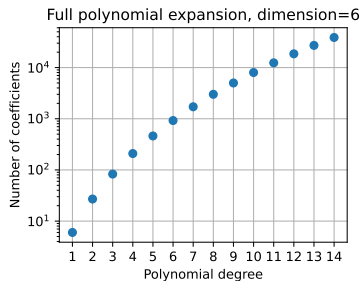


Figure 10: Number of coefficients for the polynomial chaos decomposition in dimension 6 as a function of the total degree.

⁴See [FunctionalChaosValidation](#).

Output transformation

Objective: Ensure that the metamodel output satisfies the positivity constraints imposed by physics.

For any pressure $P \in]P_{\min}, +\infty[$ and any temperature $T \in]T_{\min}, +\infty[$, we consider the transformations:

$$P_{\log} = \log(P - P_{\min}), \quad T_{\log} = \log(T - T_{\min})$$

where $P_{\min}$ is a minimum pressure and $T_{\min}$ is a minimum temperature. The inverse transformation is:

$$P = \exp(P_{\log}) + P_{\min}, \quad T = \exp(T_{\log}) + T_{\min}.$$

We use the parameters:

$$P_{\min} = 1, \quad T_{\min} = 22.85. \quad (1)$$

Polynomial chaos metamodel of the (maximum) pressure and (maximum) temperature **on the transformed variable**:

$$\tilde{P}_{\log}(z) = \sum_{k=0}^{k_{\max}^{(P)}} a_k^{(P)} \psi_k(z), \quad \tilde{T}_{\log}(z) = \sum_{k=0}^{k_{\max}^{(T)}} a_k^{(T)} \psi_k(z). \quad (2)$$

where $z \in \mathbb{R}^d$ is the standard random input vector of the model.

Results

- ▶ When the metamodel is very accurate, the Q^2 score is not relevant because it is very close to 1: it is difficult to distinguish between good and very good metamodels.
- ▶ The fraction of variance unexplained (FVU) is:

$$\text{FVU} = \frac{\text{MSE}}{\text{Var}(Y)}$$

- ▶ The Q^2 score is:

$$Q^2 = 1 - \text{FVU}$$

- ▶ To estimate the FVU, 5-fold analytical cross-validation is used⁵.

Conclusion. Optimal degree: equal to 8 for $P_{\max}$ and 8 for $T_{\max}$.

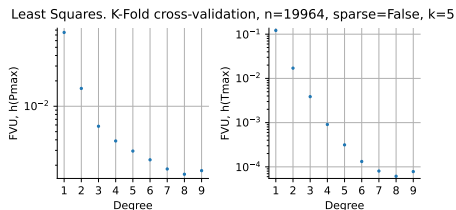


Figure 11: Fraction of variance unexplained (FVU) in the transformed space (logarithm of $P_{\max}$ and $T_{\max}$) as a function of the degree for a simple Monte Carlo design of experiments.

⁵See **FunctionalChaosValidation**.

Polynomial chaos conditional expectation

The following figure shows the sample of the logarithm of the maximum pressure as a function of each input variable, and the conditional expectation of the polynomial chaos expansion⁶.

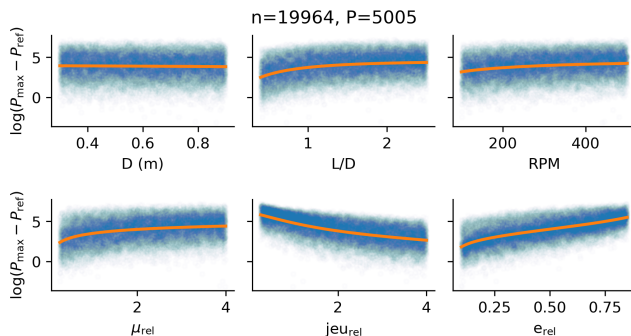


Figure 12: Monte Carlo sample of size 20,000 of the logarithm of the (shifted) variable $P_{\max}$. Conditional expectation of a polynomial chaos expansion obtained on the sample of size 20,000 with 5,005 coefficients with least squares estimation (without model selection). LEGOS calculations are performed with a dynamic tolerance (less than or equal to 10^{-4}).

⁶See `FunctionalChaosResult.getConditionalExpectation()`.

Polynomial chaos interaction Sobol' indices

The **interaction Sobol' indices** of the polynomial chaos expansion can be efficiently estimated, which helps to understand the role of the variables.

Conclusions.

- ▶ The variables e_{rel} and jeu_{rel} are highly influential on $P_{\log}$ and the variables jeu_{rel} and μ_{rel} are highly influential on $T_{\log}$.
- ▶ There is no significant interaction on pressure and little on temperature.
- ▶ Smolyak quadrature (which neglects high-order interactions) can produce good results.

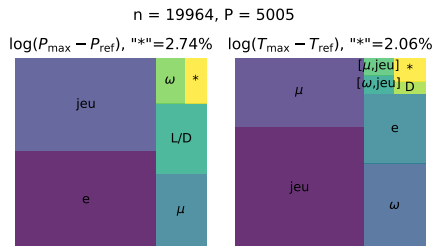


Figure 13: Interaction indices derived from the polynomial chaos expansion estimated by least squares with a sample of size 20,000 and 5,005 coefficients. Left, for the logarithm of $P_{\max}$. Right, for the logarithm of $T_{\max}$. The most influential variables are at the bottom left and the least influential are at the top right. For $P_{\max}$, less than 3% of the variance is represented by the yellow rectangle (for $T_{\max}$, less than 3% by the yellow rectangle). LEGOS calculations use a fixed tolerance of 10^{-9} .

Designs of experiments

Large-scale design of experiments

- ▶ Increasing the sample size beyond 5,000 can improve the accuracy of the metamodel.
- ▶ Target for simple Monte Carlo:
 $n = 20,000$.
- ▶ QMC Sobol': size equal to a power of 2:
 $n = 2^{14}$.
- ▶ Tensorized Gauss quadrature: uses 5 or 6 nodes per marginal.
- ▶ Sparse Smolyak quadrature: uses a level of 7 or 8.
- ▶ Development and use of an evaluation module on the supercomputer Cronos with E. Fekhari and J. Muré.²

Methods

- ▶ Quadrature methods (Gauss and Smolyak): **the sum of the weights must be equal to 1**. A “correction” of the weights is performed in case of an evaluation failure: failed points are ignored and the weights are normalized based on the successful points. This method **cannot** be effective.
- ▶ Smolyak: we would have liked to use Smolyak quadrature based on the 1D Clenshaw-Curtis quadrature, but it was not available at the time of the study.³ By default, we use Smolyak quadrature with the 1D Gauss rule.

²See [othpc](#).

³See [2843](#).

Results

Objective: Reduce the number of failed points.

Method (tol.)	n	Estim.	Degree	95% res. P.	Failures (%)
MC (fixed)	978	L.S.	5	[-16, 6] bar	1.7 %
MC (fixed)	1 964	L.S.	3	[-33, 30] bar	1.8 %
MC (fixed)	4 925	L.S.	5	[-16, 16] bar	1.5 %
MC (dyn.)	19 964	L.S.	8	[-9, 9] bar	0.18 %
QMC (dyn.)	16 363	L.S.	8	[-10, 9] bar	0.13 %
Gauss (dyn.)	15 625	Integr.	3	[-64, 253] bar	1.17 %
Gauss (dyn.)	46 075	Integr.	2	[-68, 201] bar	1.25 %
Smolyak (dyn.)	12 841	Integr.	2	[-33, 57] bar	0 %
Smolyak (dyn.)	33 193	Integr.	1	[-4284, 414] bar	0.17 %

Table 3: Accuracy of the metamodels and physical model evaluation failures on a collection of designs of experiments evaluated on Cronos. Prediction interval of the metamodel residuals. For each design of experiments, the tolerance can be fixed or dynamic. The estimation can be by least squares (“L.S.”) or by integration (“Integr.”). Legend: “fixed” means that the LEGOS tolerance is fixed (equal to 10^{-9}), “dyn” means that the LEGOS tolerance is dynamic (less than or equal to 10^{-4}).

Conclusion: Gauss (and Smolyak-Gauss) points more frequently generate LEGOS evaluation failures (points at the edges of the domain are more frequent), which defeats the Gauss and Smolyak methods. QMC fails to improve predictions compared to Monte Carlo.

Results

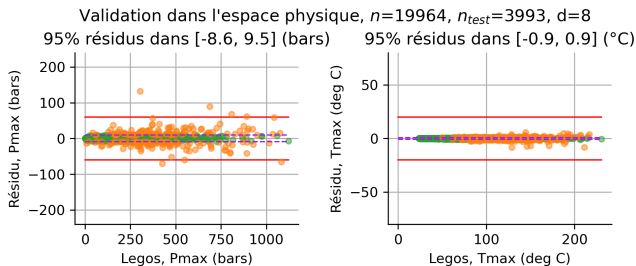


Figure 14: Metamodel residuals as a function of the LEGOS prediction. Coefficients of the polynomial chaos expansion estimated on a Monte Carlo sample of size 19,964 points with a polynomial degree equal to 8 (2×3003 coefficients). Estimation by 5-fold cross-validation.

Conclusion. With a Monte Carlo sample of size 20 000, the prediction interval is approximately ± 10 bar for $P_{\max}$ (versus ± 15 bar for $n = 5\,000$) and ± 1 °C for $T_{\max}$ (versus ± 2 °C for $n = 5\,000$).

Health indicator

We can create a turbine health indicator:

$$\mathbf{x} \rightarrow (P_{\max}, T_{\max}) \rightarrow (\text{Indicator P}, \text{Indicator T}).$$

Uncertainty propagation through the polynomial chaos metamodel combined with the health indicator produces the following results.

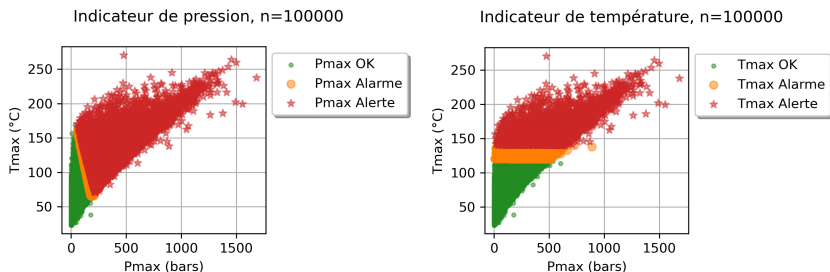


Figure 15: Value of the health indicator for a simple Monte Carlo sample of size $n = 10^5$ drawn from the distribution of the vector $(D, LsD, RPM, Mu, Clearance, Exc)$. Left, pressure indicator. Right, temperature indicator.

Conclusions and perspectives

Method:

- ▶ Polynomial chaos yields good results if the sample size is sufficient and non-convergences are reduced.

Results:

- ▶ The input domain is too large for the stability of LEGOS.
- ▶ Reducing the frequency of failures with a dynamic tolerance ($\leq 10^{-4}$) does not significantly improve prediction quality.
- ▶ Advanced experimental designs (QMC, Gauss, Smolyak) do not improve prediction accuracy over standard Monte-Carlo.
- ▶ **Suggestion:** Develop an asymptotic model in LEGOS when the oil film is very thin?

Perspectives:

1. LARS performance.

LARS model selection can be slow in OpenTURNS when the sample size is large: 2987 could improve this situation.

2. Prediction uncertainty.

Objective: provide the user with a pointwise prediction interval.
Use conformal prediction adapted to polynomial chaos.¹

3. Positive predictions.

Objective: do not use a logarithmic transformation, to avoid introducing a potential non-linearity.
Create a metamodel as a sum of squares of polynomials with nonnegative coefficients.²

¹See [Jaber et al., 2025, Loukrezis and Giovanis, 2025, Hatstatt et al., 2026].

²See [Lasserre, 2009].

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